Coulomb's Law	Biot-Savart law	Boundary Conditions	
$d\vec{E}_p = \frac{1}{4\pi\varepsilon}\frac{\rho_L\, dl}{r^2}\,\vec{a}_r$	$\vec{H} = \int_L \frac{I\,\vec{dl}\times\vec{a}_r}{4\pi r^2}$	$D_{1n} - D_{2n} = \rho_s \qquad E_{1t} = E_{2t}$	$(\vec{H}_1 - \vec{H}_2)\times\vec{a}_{n12} = \vec{K} \quad B_{1n} = B_{2n}$
$d\vec{E}_p = \frac{1}{4\pi\varepsilon}\frac{\rho_S\, dS}{r^2}\,\vec{a}_r$	$\vec{H} = \int_S \frac{\vec{K}\, ds\times\vec{a}_r}{4\pi r^2}$	$\vec{D} = \varepsilon_o\,\vec{E} + \vec{P} = \varepsilon\vec{E}$	$\vec{B} = \mu_o\,(\vec{H} + \vec{M}) = \mu\vec{H}$
$d\vec{E}_p = \frac{1}{4\pi\varepsilon}\frac{\rho_v\, dv}{r^2}\,\vec{a}_r$	$\vec{H} = \int_v \frac{\vec{J}\, dv\times\vec{a}_r}{4\pi r^2}$	$\int_V \vec{\nabla}\cdot\vec{A}\; dv = \oint_S \vec{A}\cdot\vec{dS}$	$\int_S (\vec{\nabla}\times\vec{A})\cdot\vec{dS} = \oint_L \vec{A}\cdot\vec{dl}$

Maxwell's Equations		Ampere's Law $I = \oint_l \vec{H}\cdot\vec{dl}$	$\vec{H} = \frac{I}{2\pi\rho}\vec{a}_\varphi$	Finite line $\vec{H}_p = \frac{I}{4\pi s}(\cos\alpha_2 - \cos\alpha_1)\,\vec{a}_H$
$\vec{\nabla}\times\vec{H} = \vec{J} + \frac{\partial\vec{D}}{\partial t}$	$\vec{\nabla}\cdot\vec{D} = \rho_v$	Gauss's Law $Q = \oint_s \vec{D}\cdot\vec{ds}$	$\vec{F}_m = q\,\vec{v}\times\vec{B}$	Infinite sheet $\vec{H} = (\vec{k}\times\vec{a}_n)/2$
$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$	$\vec{\nabla}\cdot\vec{B} = 0$	$\varphi = \int_s \vec{B}\cdot\vec{ds}$	$\vec{F}_m = \int_l I\,\vec{dl}\times\vec{B}$	$V_{AB} = V_B - V_A = -\int_A^B \vec{E}\,\vec{dl}$
$\vec{\nabla}\times\vec{H}_s = \vec{J}_s + j\omega\vec{D}_s$	$\vec{\nabla}\cdot\vec{D}_s = \rho_{vs}$	Faraday's Law $Emf = -\frac{\partial\varphi}{\partial t}$	$w_e = \frac{1}{2}\varepsilon E^2$	$\vec{E} = -\vec{\nabla}V \qquad \nabla^2 V = -\rho_v/\epsilon$
$\vec{\nabla}\times\vec{E}_s = -j\omega\vec{B}_s$	$\vec{\nabla}\cdot\vec{B}_s = 0$	Infinite line $\vec{H} = \frac{I}{2\pi s}\vec{a}_H$	$w_m = \frac{1}{2}\mu H^2$	$\nabla\vec{J} = -\frac{\partial\rho_v}{\partial t} \qquad \vec{J} = \sigma\vec{E}$

Rectangular Coordinates

$$\mathbf{A} = A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z$$

$$\nabla V = \frac{\partial V}{\partial x}\mathbf{a}_x + \frac{\partial V}{\partial y}\mathbf{a}_y + \frac{\partial V}{\partial z}\mathbf{a}_z$$

$$\nabla\cdot\mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla\times\mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right]\mathbf{a}_x + \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right]\mathbf{a}_y + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right]\mathbf{a}_z$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix}\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

$$d\mathbf{l} = dx\,\mathbf{a}_x + dy\,\mathbf{a}_y + dz\,\mathbf{a}_z$$

$$d\mathbf{S} = dy\,dz\,\mathbf{a}_x \quad dx\,dz\,\mathbf{a}_y \quad dx\,dy\,\mathbf{a}_z$$

$$dv = dx\,dy\,dz$$

Cylindrical Coordinates

$$\mathbf{A} = A_\rho\mathbf{a}_\rho + A_\phi\mathbf{a}_\phi + A_z\mathbf{a}_z$$

$$\nabla V = \frac{\partial V}{\partial\rho}\mathbf{a}_\rho + \frac{1}{\rho}\frac{\partial V}{\partial\phi}\mathbf{a}_\phi + \frac{\partial V}{\partial z}\mathbf{a}_z$$

$$\nabla\cdot\mathbf{A} = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho A_\rho) + \frac{1}{\rho}\frac{\partial A_\phi}{\partial\phi} + \frac{\partial A_z}{\partial z}$$

$$\nabla\times\mathbf{A} = \frac{1}{\rho}\begin{vmatrix} \mathbf{a}_\rho & \rho\mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial\rho} & \frac{\partial}{\partial\phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix} = \left[\frac{1}{\rho}\frac{\partial A_z}{\partial\phi} - \frac{\partial A_\phi}{\partial z}\right]\mathbf{a}_\rho + \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial\rho}\right]\mathbf{a}_\phi + \frac{1}{\rho}\left[\frac{\partial}{\partial\rho}(\rho A_\phi) - \frac{\partial A_\rho}{\partial\phi}\right]\mathbf{a}_z$$

$$\nabla^2 V = \frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial V}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 V}{\partial\phi^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\sin\theta & 0 \end{bmatrix}\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

$$d\mathbf{l} = d\rho\,\mathbf{a}_\rho + \rho\,d\phi\,\mathbf{a}_\phi + dz\,\mathbf{a}_z$$

$$d\mathbf{S} = \rho\,d\phi\,dz\,\mathbf{a}_\rho \quad d\rho\,dz\,\mathbf{a}_\phi \quad \rho\,d\phi\,d\rho\,\mathbf{a}_z$$

$$dv = \rho\,d\rho\,d\phi\,dz$$

Spherical Coordinates

$$\mathbf{A} = A_r\mathbf{a}_r + A_\theta\mathbf{a}_\theta + A_\phi\mathbf{a}_\phi$$

$$\nabla V = \frac{\partial V}{\partial r}\mathbf{a}_r + \frac{1}{r}\frac{\partial V}{\partial\theta}\mathbf{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial\phi}\mathbf{a}_\phi$$

$$\nabla\cdot\mathbf{A} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(A_\theta\sin\theta) + \frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial\phi}$$

$$\nabla\times\mathbf{A} = \frac{1}{r^2\sin\theta}\begin{vmatrix} \mathbf{a}_r & r\mathbf{a}_\theta & (r\sin\theta)\,\mathbf{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ A_r & rA_\theta & (r\sin\theta)\,A_\phi \end{vmatrix} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(A_\phi\sin\theta) - \frac{\partial A_\theta}{\partial\phi}\right]\mathbf{a}_r + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{\partial}{\partial r}(rA_\phi)\right]\mathbf{a}_\theta + \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial\theta}\right]\mathbf{a}_\phi$$

$$\nabla^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial V}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 V}{\partial\phi^2}$$

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ -\cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix}\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta & 0 & \cos\theta \\ \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

$$d\mathbf{l} = dr\,\mathbf{a}_r + r\,d\theta\,\mathbf{a}_\theta + r\sin\theta\,d\phi\,\mathbf{a}_\phi$$

$$d\mathbf{S} = r^2\sin\theta\,d\theta\,d\phi\,\mathbf{a}_r \quad r\sin\theta\,dr\,d\phi\,\mathbf{a}_\theta \quad r\,dr\,d\theta\,\mathbf{a}_\phi$$

$$dv = r^2\sin\theta\,dr\,d\theta\,d\phi$$